

HPMN

Smart Dimming LED Power Supply

Wiring & Control - User Manual

TVA Series Wireless smart dimming / Mi Home ecosystem	TVB Series DALI + Push DIM	TVC Series 0-10 V active dimming
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Instructions for use

Read this manual carefully before using the product and keep it for future reference. Wiring must be completed by qualified personnel in accordance with local electrical regulations.

Manufacturer: Guangdong Huapu Intelligent Technology Co., Ltd.

TVA Series - Wireless Smart Dimming

1. Features

- 1.1 Three-color switching: white, neutral white, and warm white. Each power-switch cycle changes to the next light color.
- 1.2 Stepless dimming of brightness and color temperature over the adjustment range.
- 1.3 Multiple control methods: voice control, mobile app, smart remote control, and wireless wall panel.
- 1.4 Supports two-wire, three-wire, and universal two-/three-wire connection methods.
- 1.5 Protection functions: short circuit, overload, and over-temperature.
- 1.6 Natural convection cooling.
- 1.7 Suitable for LED residential and commercial lighting applications.
- 1.8 Quick and convenient installation.
- 1.9 Designed to comply with global lighting-equipment safety requirements.
- 2.0 3-year warranty.

2. Product Specifications

Model	TVA-200-24	TVA-300-24	TVA-400-24
Rated power	200 W	300 W	400 W
Output current	8.3 A	12.5 A	16.7 A
Output voltage	24 Vdc	24 Vdc	24 Vdc
Ripple & noise	200 mV	200 mV	200 mV
Input voltage	176-264 Vac		
Input frequency	47-63 Hz		
Power factor	PF $\geq$ 0.5		
Input current	2.0 A	3.0 A	4.0 A
Efficiency	$\geq$ 86%	$\geq$ 86%	$\geq$ 86%
Overload protection	When output current exceeds the limit, output voltage decreases; automatic recovery after the load is reduced.		
Short-circuit protection	Output shuts down; automatic recovery after the fault is removed.		
Over-temperature protection	Output shuts down; automatic recovery when the temperature falls to a normal level.		
Operating temperature	-30°C to 50°C, 20-50% RH		
Storage temperature	-40°C to 85°C, 5-70% RH		
Tc	$\leq$ 75°C		
Material	Aluminum		
Lifetime	35,000 h @ Tc: 60°C		
Warranty	2 years		
Switching cycles	> 25,000		
Carton size	430 x 387 x 170 mm	488 x 375 x 165 mm	375 x 284 x 170 mm
Qty / carton	40 pcs	40 pcs	20 pcs
Product size	191 x 60.5 x 30 mm	221 x 60.5 x 30 mm	251 x 60.5 x 30 mm

TVA Series - Wiring and Smart Control

3. Wiring

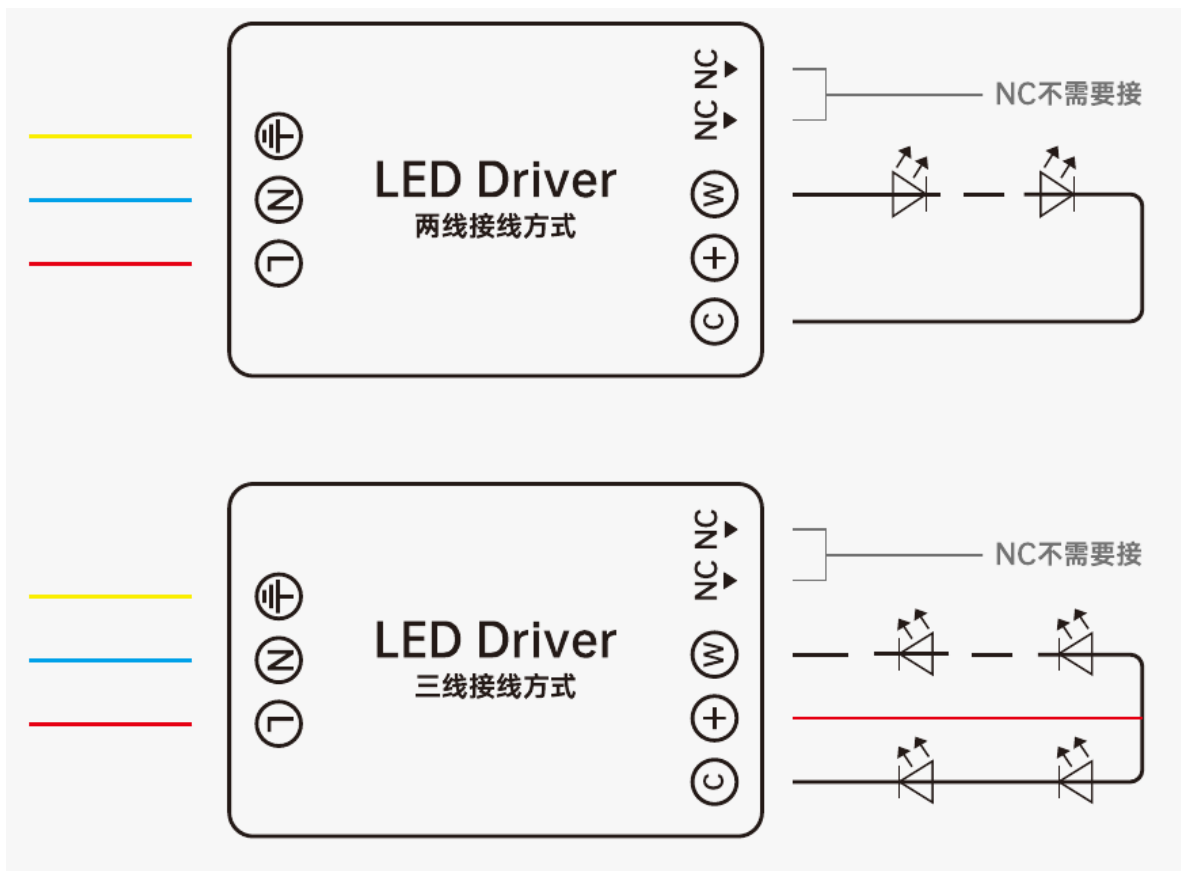


Diagram label reference: "Two-wire connection" / "Three-wire connection"; NC = no connection required.

4. Connection Methods

4.1 Xiaomi smart speaker (XiaoAI voice control)

Check that the luminaire powers on and lights normally.

Turn the luminaire on with the wall switch and say to XiaoAI: "Discover devices."

Wait briefly and follow the prompts from XiaoAI to complete setup.

Note: XiaoAI must be used with a Mesh gateway.

4.2 Mi Home App connection

Install the "Mi Home" App from the app store, register or sign in, and continue to the next step.

Tap **Add device** in the upper-right corner.

Keep the phone Bluetooth function turned on and place the phone close to the device.

Wait until network pairing is completed.

Pairing reset: The phone used for pairing must have Bluetooth enabled. If the device cannot be found, reset pairing by toggling the wall switch 8 or more times (ON for 2 s, OFF for 1 s). When the lamp flashes 5 times, it has entered pairing mode.

4.3 Wireless wall panel connection

Pairing: Within 5 seconds after power-on, press and hold the ON button until the lamp flashes.

Unpairing: Toggle the wall switch 8 or more times (ON for 2 s, OFF for 1 s). The lamp flashes 5 times when pairing information is cleared.

Capacity: One wall panel can control up to 50 devices. Effective distance: 10 m.

4.4 Smart remote-control connection

Pairing: Within 5 seconds after power-on, press and hold the SETUP button until the lamp flashes.

Unpairing: Toggle the wall switch 8 or more times (ON for 2 s, OFF for 1 s). The lamp flashes 5 times when pairing information is cleared.

Remote functions: Day mode, night mode, brightness up/down, color-temperature warmer/cooler, color-temperature switching, warm light, white light, brightness switching, ON and OFF.

Capacity: One remote control can control up to 50 devices. **Effective distance:** 10 m.

TVA Series - Operation and Troubleshooting

5. Operating Instructions

The input uses a covered 950 3-pin terminal for AC live (L), neutral (N), and protective earth. The output uses a covered 950 3-pin terminal. Connect according to the product markings and observe the correct polarity.

Important:

5.1 Distinguish the input and output terminals before wiring. Connect correctly and verify the wiring before applying power.

5.2 Connect the load to the DC output first and confirm that the wiring is correct before switching on the power. Before reconnecting the LED, wait until the energy stored at the output has been fully discharged; otherwise the LED may be damaged.

6. Abnormal Conditions and Troubleshooting

6.1 If the lamp does not light after the electrical connections are completed for the first time, disconnect the AC input and check:

6.1.1 Poor contact at the DC output terminals.

6.1.2 Whether the DC output polarity is reversed, and whether the LED board has been connected/soldered in reverse.

6.1.3 Poor contact at the AC input terminals. After correcting the above faults, test again.

6.2 If the lamp lights but flickers after installation, disconnect the AC input and check the DC output:

6.2.1 Overload (excessive load) or short circuit.

6.2.2 The power-supply design parameters do not match the actual application parameters.

If any other questions or problems occur during use, contact our company promptly and provide the relevant fault information. We will actively assist in resolving the problem.

1. Features

- 1.1 Single-channel output
- 1.2 Supports DALI dimming and touch dimming (push-button dimming).
- 1.3 Dimming range: 1-100%; smooth dimming with no flicker.
- 1.4 2-wire / 3-wire / universal 2-/3-wire connection
- 1.5 Protection functions: short circuit / overload / over-temperature
- 1.6 Natural air cooling
- 1.7 Suitable for LED residential lighting, commercial lighting, and similar applications
- 1.8 Quick and convenient installation
- 1.9 Complies with global safety requirements for lighting equipment
- 1.10 3-year warranty

2. Product Specifications

Model		TVB-200-24	TVB-300-24	TVB-400-24
OUTPUT	Rated power	200 W	300 W	400 W
	Output current	8.3 A	12.5 A	16.7 A
	Output voltage	24 Vdc	24 Vdc	24 Vdc
	Ripple & noise	200 mV	200 mV	200 mV
INPUT	Input voltage	176-264 Vac		
	Input frequency	47-63 Hz		
	Power factor	PF ≥ 0.5		
	Input current	2.0 A	3.0 A	4.0 A
	Efficiency	≥ 86%	≥ 86%	≥ 86%
PROTECTION	Overload protection	When output current exceeds the limit, output voltage decreases; automatic recovery after the load is reduced.		
	Short-circuit protection	Output shuts down; automatic recovery after the fault is removed.		
	Over-temperature protection	Output shuts down; automatic recovery when the temperature falls to a normal level.		
OTHER	Operating temperature	-30°C to 50°C, 20-50% RH		
	Storage temperature	-40°C to 85°C, 5-70% RH		
	Tc	≤ 75°C		
	Material	Aluminum		
	Lifetime	35,000 h @ Tc: 80°C		
	Warranty period	2 years		
	Switching cycles	> 25,000		
	Carton size	430 x 387 x 170 mm	488 x 375 x 165 mm	375 x 284 x 170 mm
	Qty / carton	40 pcs	40 pcs	20 pcs
	Product size	191 x 60.5 x 30 mm	221 x 60.5 x 30 mm	251 x 60.5 x 30 mm

3. Wiring

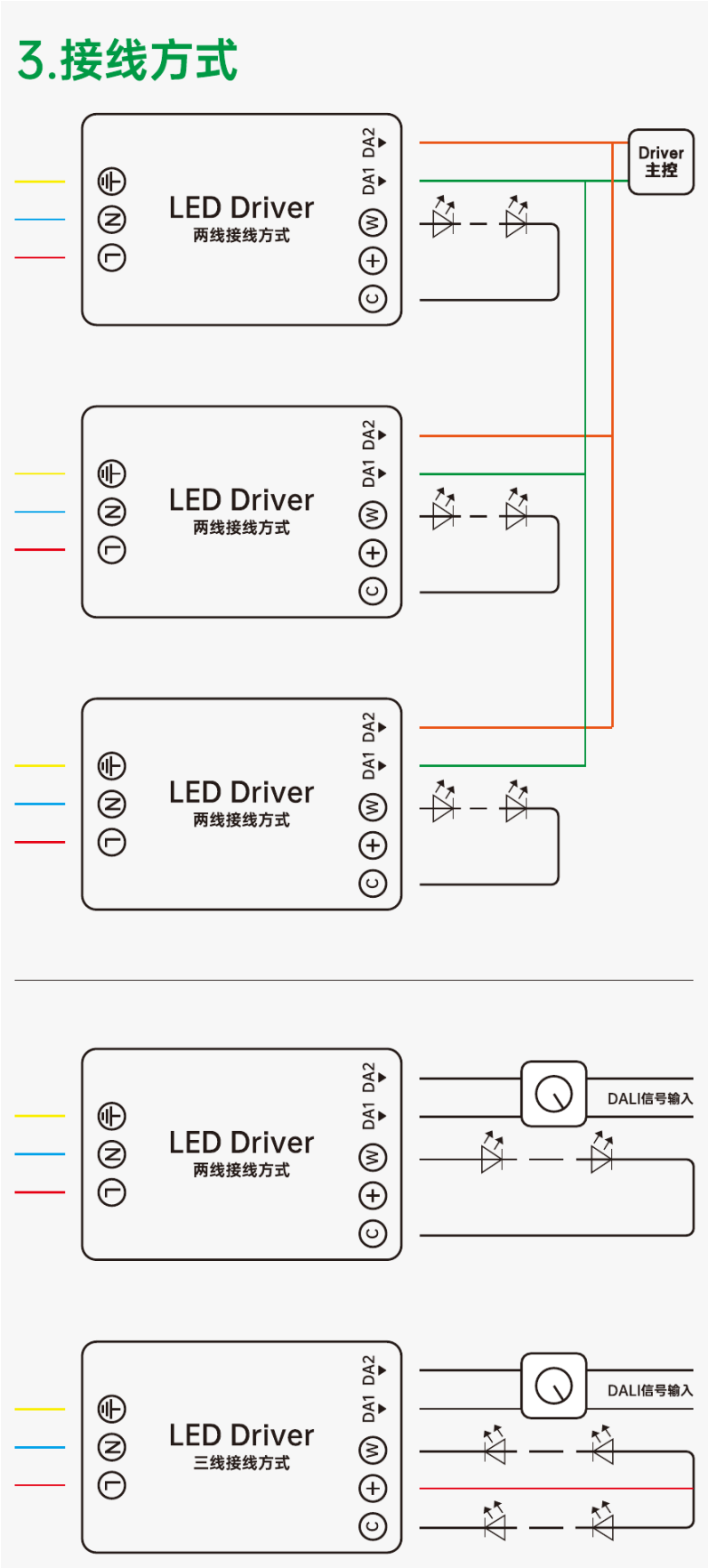
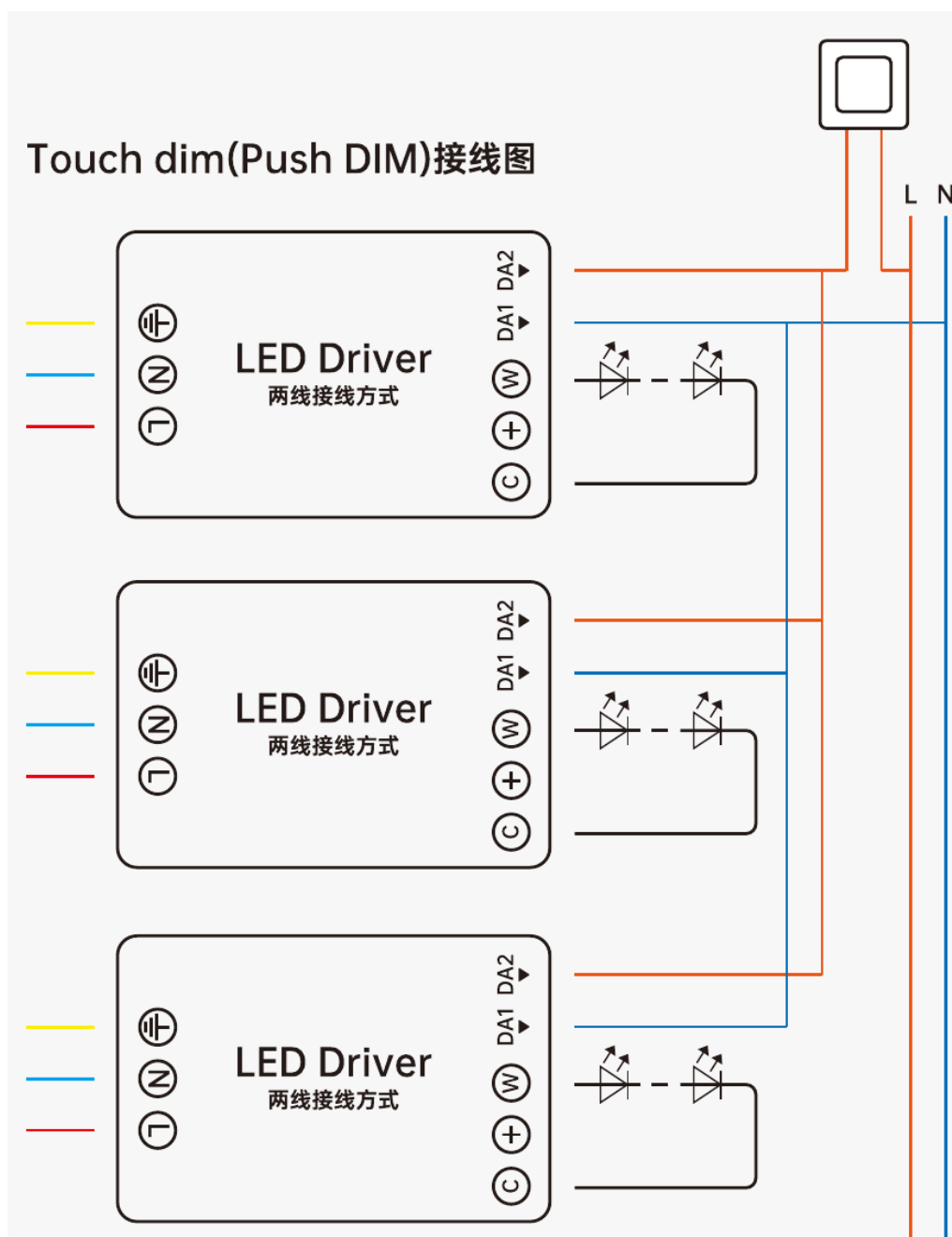


Diagram label reference: Driver/master controller; DALI signal input; two-wire connection; three-wire connection.

TVB Series - Touch DIM / Push DIM Wiring



Touch DIM / Push DIM: the push switch is connected as shown in the original wiring diagram.

TVB Series - Push DIM, Operation and Troubleshooting

3.1 Push DIM function

3.1.1 Short-press the Push DIM (Touch DIM) switch for less than 0.5 s to switch the lamp between on and off.

3.1.2 Press and hold the Push DIM (Touch DIM) switch for 1-3 s to adjust brightness. The dimming direction alternates with each press (dim up / dim down).

3.1.3 Press and hold the Push DIM (Touch DIM) switch for 10 s so that all connected luminaires reach the same brightness level (50%).

3.1.4 Brightness adjustment range: 1-100%.

3.1.5 Power-off memory: after power is restored, the driver returns to the state before the previous power interruption.

4. Operating Instructions

The input uses a covered HB950 3-pin terminal for AC live (L), neutral (N), and protective earth. The output uses a covered HB950 3-pin terminal, and the signal input uses a DA260-500-2P terminal. Connect according to the product markings and observe the correct polarity.

Important:

4.1 Distinguish the input and output terminals before wiring. Connect correctly and verify the wiring before applying power.

4.2 Connect the load to the DC output first and confirm the wiring is correct before switching on the power. Before reconnecting the LED, wait until the energy stored at the output has been fully discharged; otherwise the LED may be damaged.

5. Abnormal Conditions and Troubleshooting

5.1 If the lamp does not light after the electrical connections are completed for the first time, disconnect the AC input and check:

5.1.1 Poor contact at the DC output terminals.

5.1.2 Whether the DC output polarity is reversed, and whether the LED board has been connected/soldered in reverse.

5.1.3 Poor contact at the AC input terminals. After correcting the above faults, test again.

5.2 If the lamp lights but flickers after installation, disconnect the AC input and check the DC output:

5.2.1 Overload (excessive load) or short circuit.

5.2.2 The power-supply design parameters do not match the actual application parameters.

If any other questions or problems occur during use, contact our company promptly and provide the relevant fault information. We will actively assist in resolving the problem.

TVC Series - 0-10 V Dimming

1. Features

- 1.1 Single-channel constant-voltage output.
- 1.2 Supports active 0-10 V analog dimming.
- 1.3 Dimming range: 1-100%; smooth dimming with no visible flicker.
- 1.4 Protection functions: short circuit, overload, and over-temperature.
- 1.5 Natural convection cooling.
- 1.6 Suitable for LED residential and commercial lighting applications.
- 1.7 Quick and convenient installation.
- 1.8 Designed to comply with global lighting-equipment safety requirements.
- 1.9 3-year warranty.

2. Product Specifications

Model	TVC-200-24	TVC-300-24	TVC-400-24
Rated power	200 W	300 W	400 W
Output current	8.3 A	12.5 A	16.7 A
Output voltage	24 Vdc	24 Vdc	24 Vdc
Ripple & noise	200 mV	200 mV	200 mV
Input voltage	176-264 Vac		
Input frequency	47-63 Hz		
Power factor	PF ≥ 0.5		
Input current	2.0 A	3.0 A	4.0 A
Efficiency	≥ 86%	≥ 86%	≥ 86%
Overload protection	When output current exceeds the limit, output voltage decreases; automatic recovery after the load is reduced.		
Short-circuit protection	Output shuts down; automatic recovery after the fault is removed.		
Over-temperature protection	Output shuts down; automatic recovery when the temperature falls to a normal level.		
Operating temperature	-30°C to 50°C, 20-50% RH		
Storage temperature	-40°C to 85°C, 5-70% RH		
Tc	≤ 75°C		
Material	Aluminum		
Lifetime	35,000 h @ Tc: 80°C		
Warranty	2 years		
Switching cycles	> 25,000		
Carton size	430 x 387 x 170 mm	488 x 375 x 165 mm	375 x 284 x 170 mm
Qty / carton	40 pcs	40 pcs	20 pcs
Product size	191 x 60.5 x 30 mm	221 x 60.5 x 30 mm	251 x 60.5 x 30 mm

TVC Series - Wiring, Operation and Troubleshooting

3. Wiring

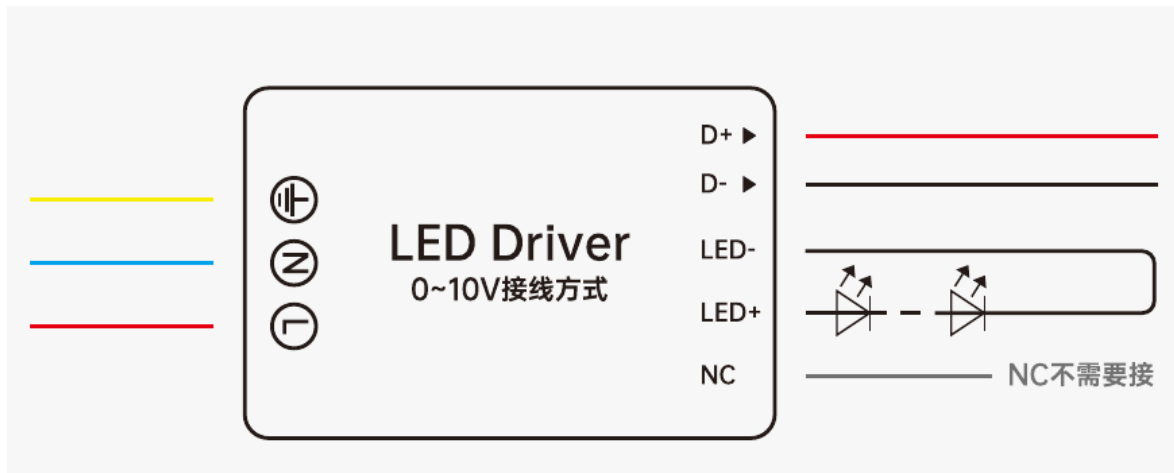


Diagram label reference: D+ / D- = 0-10 V dimming signal; LED+ / LED- = LED output; NC = no connection required.

4. Operating Instructions

The input uses a covered HB950 3-pin terminal for AC live (L), neutral (N), and protective earth. The output uses a covered HB950 3-pin terminal, and the signal input uses a DA260-500-2P terminal. Connect according to the product markings and observe the correct polarity.

Important:

- 4.1** Distinguish the input and output terminals before wiring. Connect correctly and verify the wiring before applying power.
- 4.2** Connect the load to the DC output first and confirm the wiring is correct before switching on the power. Before reconnecting the LED, wait until the energy stored at the output has been fully discharged; otherwise the LED may be damaged.

5. Abnormal Conditions and Troubleshooting

5.1 If the lamp does not light after the electrical connections are completed for the first time, disconnect the AC input and check:

- 5.1.1** Poor contact at the DC output terminals.
- 5.1.2** Whether the DC output polarity is reversed, and whether the LED board has been connected/soldered in reverse.
- 5.1.3** Poor contact at the AC input terminals. After correcting the above faults, test again.

5.2 If the lamp lights but flickers after installation, disconnect the AC input and check the DC output:

- 5.2.1** Overload (excessive load) or short circuit.
- 5.2.2** The power-supply design parameters do not match the actual application parameters.

If any other questions or problems occur during use, contact our company promptly and provide the relevant fault information. We will actively assist in resolving the problem.